

Advanced Deep Learning - Exercise 3

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1 Comparing ViT and CNN Architectures

a) Input Processing - ViT breaks down the input image into non-overlapping patches at first, and then they are flattened along with their position embeddings and class labels, before inputting to the Transformer Encoder.

In a standard CNN, the features are learnt with exponentially increasing receptive fields (along the depth of the network), by primarily leveraging the translational invariance and local inference properties of the architecture. Unlike in transformers, locality is inferred by design in CNNs, while ViT needs to rely on data to learn the spatial relationships.

b) Data Efficiency - As a result of this model inductive bias of CNNs, they learn spatial relationships like translational invariance and locality easily. While ViT takes significantly higher data to achieve this, this also helps them learn long-distance (spatially) correlations well.

c) Complexity Analysis - For an image with N patches, global self attention in ViT has quadratic complexity - $O(N^2)$. This can prove to be a bottle neck in time, but more importantly memory, for images with high resolution. Where as in CNNs, the complexity scales only linearly.

2 DETR Training: Forward Pass & Loss Calculation

a) Qualitative Analysis - $\hat{y}_4$ seems to be the hallucination. $\hat{y}_1$ is the prediction that correctly predicts the spatial coordinates of y_1 (Eagle) but classifies it incorrectly. $\hat{y}_2$ is the prediction that classifies Eagle (y_1) correctly, but spatially inaccurate.

b) Bipartite Matching -
Class Probabilities -

Prediction	Eagle (y_1)	Pigeon	Crow	Owl (y_2)	No Object
$\hat{y}_1$	0.083	0.614	0.083	0.083	0.137
$\hat{y}_2$	0.614	0.083	0.083	0.083	0.137
$\hat{y}_3$	0.083	0.083	0.083	0.614	0.137
$\hat{y}_4$	0.083	0.614	0.083	0.083	0.137

Box Distances -

Prediction	y_1	y_2
$\hat{y}_1$	0	5.66
$\hat{y}_2$	1.41	4.24
$\hat{y}_3$	5.66	0
$\hat{y}_4$	11.31	5.66

Cost matrix (Cost for both the appended No object ground truths is 0, by definition) -

Prediction	y1	y2
$\hat{y}1$	$0 - 0.083 = -0.083$	$5.66 - 0.083 = 5.577$
$\hat{y}2$	$1.41 - 0.614 = 0.796$	$4.24 - 0.083 = 4.157$
$\hat{y}3$	$5.66 - 0.083 = 5.577$	$0 - 0.614 = -0.614$
$\hat{y}4$	$11.31 - 0.083 = 11.227$	$5.66 - 0.083 = 5.577$

Final Prediction Assignments - After analyzing the 12 possible assignments, assigning y2 with $\hat{y}3$ and y1 with $\hat{y}1$ gives the optimal assignment.

Prediction	Ground Truth Assignment
$\hat{y}1$	y1 (Eagle)
$\hat{y}2$	No Object
$\hat{y}3$	y2 (Owl)
$\hat{y}4$	No Object

c) Total Training Loss - Hungarian Loss = $(-0.083) + (-0.614) + (-0.137) + (-0.137) = -0.971$